

Mini Project -1

Develop a movie recommendation model using the scikit-learn library in python.
Refer dataset

https://github.com/rashida048/Some-NLP-Projects/blob/master/movie_dataset.csv

```
In [1]: from sklearn.metrics.pairwise import cosine_similarity
import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

```
In [3]: df = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\MP1\movie_dataset.csv")
```

```
In [4]: df.head()
```

Out[4]:

	index	budget	genres	homepage	id	keywords	original_language
0	0	237000000	Action Adventure Fantasy Science Fiction	http://www.avatarmovie.com/	19995	culture clash future space war space colony so...	
1	1	300000000	Adventure Fantasy Action	http://disney.go.com/disneypictures/pirates/	285	ocean drug abuse exotic island east india trad...	
2	2	245000000	Action Adventure Crime	http://www.sonypictures.com/movies/spectre/	206647	spy based on novel secret agent sequel mi6	
3	3	250000000	Action Crime Drama Thriller	http://www.thedarkknighttrises.com/	49026	dc comics crime fighter terrorist secret ident...	
4	4	260000000	Action Adventure Science Fiction	http://movies.disney.com/john-carter	49529	based on novel mars medallion space travel pri...	

5 rows × 24 columns

```
In [5]: features = ['keywords', 'cast', 'genres', 'director']
```

```
In [6]: def combine_features(row):
return row['keywords']+" "+row['cast']+" "+row['genres']+" "+row['director']
```

```
In [7]: for feature in features:
        df[feature] = df[feature].fillna('')

        df["combined_features"] = df.apply(combine_features,axis=1)
```

```
In [8]: cv = CountVectorizer()
        count_matrix = cv.fit_transform(df["combined_features"])
```

```
In [9]: cosine_sim = cosine_similarity(count_matrix)
```

```
In [10]: def get_title_from_index(index):
          return df[df.index == index]["title"].values[0]
          def get_index_from_title(title):
              return df[df.title == title]["index"].values[0]
```

```
In [11]: movie_user_likes = "Avatar"
          movie_index = get_index_from_title(movie_user_likes)
          similar_movies = list(enumerate(cosine_sim[movie_index]))
```

```
In [12]: sorted_similar_movies = sorted(similar_movies,key=lambda x:x[1],reverse=True)[1:]
```

```
In [13]: i=0
          print("Top 5 similar movies to "+movie_user_likes+" are:\n")
          for element in sorted_similar_movies:
              print(get_title_from_index(element[0]))
              i=i+1
              if i>5:
                  break
```

Top 5 similar movies to Avatar are:

Guardians of the Galaxy
Aliens
Star Wars: Clone Wars: Volume 1
Star Trek Into Darkness
Star Trek Beyond
Alien

Mini Project - 2

Use the following covid_vaccine_statewise.csv dataset and perform following analytics on the given dataset

[https://www.kaggle.com/sudalairajkumar/covid19-in-india?](https://www.kaggle.com/sudalairajkumar/covid19-in-india?select=covid_vaccine_statewise.csv)

[select=covid_vaccine_statewise.csv](#)

- Describe the dataset
- Number of persons state wise vaccinated for first dose in India
- Number of persons state wise vaccinated for second dose in India
- Number of Males vaccinated
- Number of females vaccinated

```
In [38]: import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import seaborn as sns
```

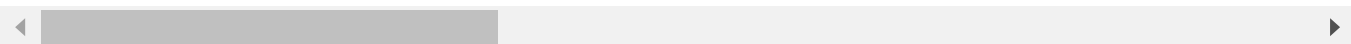
```
In [58]: vaccine_data = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\MP2\covid_vaccine_stat
```

```
In [59]: vaccine_data.head(5)
```

Out[59]:

	Updated On	State	Total Doses Administered	Sessions	Sites	First Dose Administered	Second Dose Administered	Male (Doses Administered)	Female (Doses Administered)
0	16/01/2021	India	48276.0	3455.0	2957.0	48276.0	0.0	NaN	NaN
1	17/01/2021	India	58604.0	8532.0	4954.0	58604.0	0.0	NaN	NaN
2	18/01/2021	India	99449.0	13611.0	6583.0	99449.0	0.0	NaN	NaN
3	19/01/2021	India	195525.0	17855.0	7951.0	195525.0	0.0	NaN	NaN
4	20/01/2021	India	251280.0	25472.0	10504.0	251280.0	0.0	NaN	NaN

5 rows × 10 columns

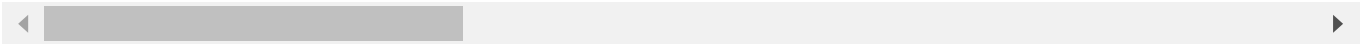


In [60]: vaccine_data.describe()

Out[60]:

	Total Doses Administered	Sessions	Sites	First Dose Administered	Second Dose Administered	Male (Doses Administered)	Female (Doses Administered)
count	7.621000e+03	7.621000e+03	7621.000000	7.621000e+03	7.621000e+03	7.461000e+03	7.461000e+03
mean	9.188171e+06	4.792358e+05	2282.872064	7.414415e+06	1.773755e+06	3.620156e+06	3.168416e+06
std	3.746180e+07	1.911511e+06	7275.973730	2.995209e+07	7.570382e+06	1.737938e+07	1.515310e+07
min	7.000000e+00	0.000000e+00	0.000000	7.000000e+00	0.000000e+00	0.000000e+00	2.000000e+00
25%	1.356570e+05	6.004000e+03	69.000000	1.166320e+05	1.283100e+04	5.655500e+04	5.210700e+04
50%	8.182020e+05	4.547000e+04	597.000000	6.614590e+05	1.388180e+05	3.897850e+05	3.342380e+05
75%	6.625243e+06	3.428690e+05	1708.000000	5.387805e+06	1.166434e+06	2.735777e+06	2.561513e+06
max	5.132284e+08	3.501031e+07	73933.000000	4.001504e+08	1.130780e+08	2.701636e+08	2.395186e+08

8 rows × 22 columns



```
In [61]: dose1 = vaccine_data.groupby("State")[["First Dose Administered"]].max()  
dose1
```

Out[61]:

First Dose Administered	
State	
Andaman and Nicobar Islands	216046.0
Andhra Pradesh	17628583.0
Arunachal Pradesh	692475.0
Assam	10495293.0
Bihar	23350171.0
Chandigarh	700285.0
Chhattisgarh	9181482.0
Dadra and Nagar Haveli and Daman and Diu	584370.0
Delhi	7835546.0
Goa	1094392.0
Gujarat	28101222.0
Haryana	10086831.0
Himachal Pradesh	4249849.0
India	400150406.0
Jammu and Kashmir	5318516.0
Jharkhand	8382280.0
Karnataka	25847691.0
Kerala	15670747.0
Ladakh	188699.0
Lakshadweep	51156.0
Madhya Pradesh	29723036.0
Maharashtra	35040812.0
Manipur	1159424.0
Meghalaya	938572.0
Mizoram	654946.0
Nagaland	632120.0
Odisha	13954592.0
Puducherry	601591.0
Punjab	8005636.0
Rajasthan	27008606.0
Sikkim	497851.0
Tamil Nadu	20836674.0
Telangana	11649268.0
Tripura	2411195.0
Uttar Pradesh	45932488.0
Uttarakhand	5070544.0

First Dose Administered

State

West Bengal	23257417.0
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```
In [62]: dose2 = vaccine_data.groupby("State")[["Second Dose Administered"]].max()  
dose2
```

Out[62]:

Second Dose Administered	
State	
Andaman and Nicobar Islands	94597.0
Andhra Pradesh	6214312.0
Arunachal Pradesh	186619.0
Assam	2208577.0
Bihar	4484768.0
Chandigarh	223534.0
Chhattisgarh	2587695.0
Dadra and Nagar Haveli and Daman and Diu	80851.0
Delhi	3000536.0
Goa	302519.0
Gujarat	9051153.0
Haryana	2923550.0
Himachal Pradesh	1382592.0
India	113077994.0
Jammu and Kashmir	1489826.0
Jharkhand	1996014.0
Karnataka	7432852.0
Kerala	6426984.0
Ladakh	70337.0
Lakshadweep	17139.0
Madhya Pradesh	5733640.0
Maharashtra	12112554.0
Manipur	246694.0
Meghalaya	231982.0
Mizoram	206773.0
Nagaland	159388.0
Odisha	4200094.0
Puducherry	151771.0
Punjab	2285629.0
Rajasthan	8375056.0
Sikkim	151538.0
Tamil Nadu	4686034.0
Telangana	3965624.0
Tripura	804099.0
Uttar Pradesh	8515236.0
Uttarakhand	1596572.0

Second Dose Administered

State

West Bengal	9132961.0
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```
In [63]: male = vaccine_data.groupby("State")[["Male(Individuals Vaccinated)"]].max()  
male
```

Out[63]:

Male(Individuals Vaccinated)	
State	
Andaman and Nicobar Islands	NaN
Andhra Pradesh	NaN
Arunachal Pradesh	NaN
Assam	NaN
Bihar	NaN
Chandigarh	NaN
Chhattisgarh	NaN
Dadra and Nagar Haveli and Daman and Diu	NaN
Delhi	NaN
Goa	NaN
Gujarat	NaN
Haryana	NaN
Himachal Pradesh	NaN
India	134941971.0
Jammu and Kashmir	NaN
Jharkhand	NaN
Karnataka	NaN
Kerala	NaN
Ladakh	NaN
Lakshadweep	NaN
Madhya Pradesh	NaN
Maharashtra	NaN
Manipur	NaN
Meghalaya	NaN
Mizoram	NaN
Nagaland	NaN
Odisha	NaN
Puducherry	NaN
Punjab	NaN
Rajasthan	NaN
Sikkim	NaN
Tamil Nadu	NaN
Telangana	NaN
Tripura	NaN
Uttar Pradesh	NaN
Uttarakhand	NaN

Male(Individuals Vaccinated)

State

West Bengal	NaN
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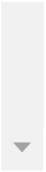
```
In [64]: female = vaccine_data.groupby("State")["Female(Individuals Vaccinated)"].max()  
female
```

Out[64]:

Female(Individuals Vaccinated)	
State	
Andaman and Nicobar Islands	NaN
Andhra Pradesh	NaN
Arunachal Pradesh	NaN
Assam	NaN
Bihar	NaN
Chandigarh	NaN
Chhattisgarh	NaN
Dadra and Nagar Haveli and Daman and Diu	NaN
Delhi	NaN
Goa	NaN
Gujarat	NaN
Haryana	NaN
Himachal Pradesh	NaN
India	115668447.0
Jammu and Kashmir	NaN
Jharkhand	NaN
Karnataka	NaN
Kerala	NaN
Ladakh	NaN
Lakshadweep	NaN
Madhya Pradesh	NaN
Maharashtra	NaN
Manipur	NaN
Meghalaya	NaN
Mizoram	NaN
Nagaland	NaN
Odisha	NaN
Puducherry	NaN
Punjab	NaN
Rajasthan	NaN
Sikkim	NaN
Tamil Nadu	NaN
Telangana	NaN
Tripura	NaN
Uttar Pradesh	NaN
Uttarakhand	NaN

State

West Bengal	NaN
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```
In [65]: trans = vaccine_data.groupby("State")[["Transgender(Individuals Vaccinated)"]].max()  
trans
```

Out[65]:

Transgender(Individuals Vaccinated)	
State	
Andaman and Nicobar Islands	NaN
Andhra Pradesh	NaN
Arunachal Pradesh	NaN
Assam	NaN
Bihar	NaN
Chandigarh	NaN
Chhattisgarh	NaN
Dadra and Nagar Haveli and Daman and Diu	NaN
Delhi	NaN
Goa	NaN
Gujarat	NaN
Haryana	NaN
Himachal Pradesh	NaN
India	46462.0
Jammu and Kashmir	NaN
Jharkhand	NaN
Karnataka	NaN
Kerala	NaN
Ladakh	NaN
Lakshadweep	NaN
Madhya Pradesh	NaN
Maharashtra	NaN
Manipur	NaN
Meghalaya	NaN
Mizoram	NaN
Nagaland	NaN
Odisha	NaN
Puducherry	NaN
Punjab	NaN
Rajasthan	NaN
Sikkim	NaN
Tamil Nadu	NaN
Telangana	NaN
Tripura	NaN
Uttar Pradesh	NaN
Uttarakhand	NaN

State

West Bengal

NaN

In []: